# Ryan McWay

## Research Statement

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### Research Agenda

My research asks whether education reforms that work in pilots can deliver sustained learning gains when governments implement them at scale, and what determines whether they do. I pair causal inference on administrative and original assessment data with economic structure: I estimate policy-relevant effects credibly, and where possible link them to a model that explains why effects persist, attenuate, or grow. My work evaluates reforms and technologies in Morocco, and Jordan, and shares a common goal: producing credible evidence that helps governments allocate scarce resources toward foundational learning.

### Job Market Paper: Scaling Education Reform in Morocco

**Question and context.** In *Scaling Education Reform: Evidence from Morocco's Pioneer School Program*, I ask whether large learning gains survive the move from pilot to national scale. Morocco ranked second to last of 57 countries on the 2021 PIRLS reading assessment, and 59 percent of fourth graders did not reach the minimum benchmark. The Pioneer School Program (PSP) is the government's core basic-education reform. It combines two "great buy" approaches identified by the Global Education Evidence Advisory Panel, structured pedagogy and targeted remediation by learning level, and delivers them through ministry structures including teacher specialization, inspector-led coaching, and school certification. The rollout expanded from 626 schools in 2023–24 to 4,626 in 2025–26, which lets me compare effects across rollout cohorts and years of exposure.

**Design.** Treatment is not randomly assigned, so I use a matched-pair difference-in-differences design. Administrative records on all public primary schools (roughly 24 million student records over 2015–2022) feed a post-double-selection LASSO that picks 16 predictors from 248 school variables for a calibrated Mahalanobis matching procedure. Original baseline and endline assessments in Arabic, French, and mathematics, linked across grades and rounds with item response theory, cover three cohorts, 483 schools, and over 36,000 students. Seven years of pre-program exam data support parallel trends, and standard errors are clustered at the matched-pair level.

**Findings.** PSP produces large gains in foundational learning. Pooling cohorts, one year of exposure raises achievement by 0.60 standard deviations (SD) across the three subjects, and two years raises it by 0.61 SD. The first cohort gains 0.90 SD after one year and 0.62 SD after two years; after three years it retains 0.53 SD, driven by French (0.94 SD) and mathematics (0.54 SD). The two-year French effect lies above the 99th percentile, and the mathematics effect near the 95th percentile, of the 96 developing-country evaluations compiled by Evans and Yuan (2022). Gains are slightly larger for girls and for students in the bottom quartile of baseline performance, though differences are modest.

| **Overall ITT effect (SD)** | **1 year** | **2 years** | **3 years** |
|---|---|---|---|
| Cohort 1 (adopted 2023) | 0.90 | 0.62 | 0.53 |
| Cohort 2 (adopted 2024) | 0.51 | 0.58 | – |
| Cohort 3 (adopted 2025) | 0.40 | – | – |

**Scaling interpretation.** Effects are large but dissipate: they fall with exposure for the first cohort and are smaller for later cohorts, and a stable-sample analysis shows this is not an artifact of schools leaving the sample. Rather than a binary "works or does not work" verdict, I organize these dynamics with the voltage-effect framework (List, 2026), which separates implementation quality, sample composition, and effective dosage. The program changed during scale-up in ways that map onto these channels: intensive remediation was shortened from six to four weeks, inspector training and school visits became less intensive, the PSP-specific teacher bonus was folded into a national salary adjustment, and later cohorts include more rural and higher-poverty schools. I treat these as suggestive explanations because available data cannot test them directly.

**Contribution.** The paper (i) provides evidence on a large government-led reform that bundles pedagogical and organizational change, (ii) shows how machine-learning-assisted matching and difference-in-differences can evaluate a non-random national rollout with evolving samples, and (iii) links reduced-form estimates to a structural scaling framework that connects short-run effect sizes to the question of whether reforms hold up at national scale.

## Related Work

**Technology-delivered personalization in Jordan.** In *Impact Evaluation of the Darsel Math Personalized Learning Platform in Jordan* (with Caroline Krafft, Paul Glewwe, Tessa Bold, Abdulhamid Haidar, Paola Giannattasio, and Mojtaba Mahdipour; *Journal of Development Economics*, Stage 1 conditionally accepted), we conduct a pre-registered randomized trial of an AI-enabled math chatbot, delivered over low-bandwidth messaging, that adaptively identifies and remediates learning gaps. We study grade-6 students in Jordan, where enrollment is high but learning levels remain low, and measure effects on mathematics as well as math self-efficacy, math anxiety, and liking of math. Preliminary estimates suggest that the introduction of Darsel improved math learning by 0.13 standard deviations. The project is a natural companion to the Morocco study: both operationalize teaching at the right level, one through teacher-delivered remediation inside a national reform and the other through scalable personalization in government schools.

**Credibility of the evidence.** In *Outsourcing Education: Experimental Evidence from Liberia* (replication of Romero, Sandefur, and Sandholtz, 2020), I reproduce and stress-test a major public-private-partnership experiment using a multiverse framework with 320 alternative specifications. The findings support positive effects on learning but show that magnitudes are sensitive to some analytic choices. This work reflects my emphasis on both impact estimation and the credibility of the evidence used for policy.

## Ongoing and Future Work

**Structural synthesis of EdTech evidence.** I am building a large synthesis of education-technology randomized evaluations drawn from repositories such as J-PAL and Harvard Dataverse. The goal is to estimate a structural model of student learning that nests heterogeneity in EdTech design (in-school versus at-home delivery, adaptive versus fixed content, game-based versus test-based interfaces) and allows teachers and parents to respond as complements or substitutes. This moves beyond average effects to the conditions under which EdTech amplifies, crowds out, or reallocates other educational inputs.